

NEW HAVEN TWEED, CT, USA

WMO: 725045

Lat: 41.264N

Lon: 72.887W

Elev: 3

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 14758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.6	(c) 14.2	(d) -7.8	(e) 3.6	(f) 12.8	(g) -3.0	(h) 4.7	(i) 17.8	(j) 24.8	(k) 29.5	(l) 20.9	(m) 31.6	(n) 8.9	(o) 320	(p) 0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	87.5	74.3	85.0	72.8	82.8	71.6	76.9	83.9	75.5	81.5	74.3	79.9	7.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	130.5	80.7	73.4	124.0	78.8	72.4	119.8	78.1	40.2	84.1	38.9	81.8	37.7	79.7	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.7	(b) 17.2	(c) 15.5	(e) 3.5	(f) 92.2	(g) 5.4	(h) 3.6	(i) -0.3	(j) 94.8	(k) -3.5	(l) 97.0	(m) -6.6	(n) 99.0	(o) -10.5	(p) 101.6		
(a) 19.7	(b) 17.2	(c) 15.5	(e) 2.1	(f) 78.9	(g) 5.1	(h) 1.4	(i) -1.6	(j) 80.0	(k) -4.6	(l) 80.8	(m) -7.5	(n) 81.6	(o) -11.2	(p) 82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	30.5	32.1	38.7	49.2	58.9	68.1	74.7	73.4	66.9	55.7	45.2	36.6
	DBStd	16.91	9.25	8.18	7.94	6.93	6.37	5.81	4.60	4.98	5.98	7.78	7.20	7.99
	HDD50	2195	606	503	357	94	5	0	0	0	0	33	178	419
	HDD65	5355	1070	922	815	474	211	35	1	4	48	302	594	879
	CDD50	3150	0	0	7	72	280	543	764	726	508	211	34	5
	CDD65	836	0	0	0	1	21	128	301	264	106	15	0	0
	CDH74	5425	0	0	0	19	114	701	2208	1837	517	29	0	0
	CDH80	1232	0	0	0	2	15	141	594	406	71	3	0	0
(14) Wind	WSAvg	6.9	7.7	7.6	8.0	7.6	6.6	5.9	5.5	5.7	6.6	7.1	7.3	7.5
(15) Precipitation	PrecAvg	41.9	2.9	3.1	3.5	3.9	3.9	3.5	3.5	3.1	3.4	3.5	4.0	3.8
	PrecMax	57.4	6.1	5.3	9.2	10.7	10.6	12.7	8.7	7.8	7.7	8.3	8.7	7.7
	PrecMin	27.7	0.2	0.6	0.7	1.4	0.8	0.2	0.8	0.7	0.9	0.3	0.4	1.0
	PrecStd	8.3	1.6	1.1	1.8	2.0	2.4	2.8	2.1	1.8	2.0	2.0	2.3	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	55.8	63.8	76.5	81.6	87.8	92.1	89.5	85.7	77.5	66.1	60.0
		MCWB	50.2	48.7	53.9	60.0	66.5	73.1	75.7	75.8	74.4	68.2	60.9	54.9
	2%	DB	50.4	50.8	57.6	69.2	77.1	83.9	88.4	86.4	81.9	73.3	63.0	55.4
		MCWB	47.4	45.8	48.9	56.2	65.6	70.4	74.5	74.3	70.9	67.9	57.8	52.2
	5%	DB	46.6	47.2	53.8	64.5	73.2	80.9	85.7	84.2	79.2	70.9	60.5	52.5
		MCWB	42.9	42.3	46.8	53.4	62.8	69.1	73.3	72.9	69.7	65.4	56.1	49.6
	10%	DB	44.0	43.9	50.3	60.6	69.9	78.1	83.2	82.1	76.9	68.0	57.7	49.6
		MCWB	40.2	39.0	44.1	51.2	60.4	67.4	72.1	72.0	68.8	62.3	53.2	45.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	51.7	56.6	63.1	70.4	75.1	78.7	78.9	76.5	71.7	63.2	56.9
		MCDB	53.6	53.0	61.3	72.8	78.5	84.7	88.7	86.0	82.5	74.2	65.9	58.1
	2%	WB	48.0	46.7	51.5	58.9	67.7	73.1	76.9	77.2	74.4	69.3	59.7	53.1
		MCDB	49.7	49.8	55.2	65.1	74.6	80.6	85.1	83.3	79.0	72.8	62.0	55.0
	5%	WB	43.9	43.1	48.0	56.1	65.2	71.4	75.4	75.8	72.7	66.5	57.1	50.4
		MCDB	46.2	46.5	52.0	61.7	71.2	77.8	82.1	81.1	76.7	69.9	59.6	52.5
	10%	WB	40.6	39.7	45.4	53.2	62.6	69.7	74.3	74.5	70.9	63.7	54.3	46.2
		MCDB	43.6	43.3	49.3	58.0	67.7	75.1	80.5	79.4	74.9	67.3	57.2	48.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	14.8	15.0	16.6	16.1	15.3	15.2	15.1	16.2	15.8	15.5	13.1
		MCDBR	16.6	18.5	19.9	23.2	21.1	18.4	17.4	16.3	17.4	16.4	17.3	15.5
		MCWBR	15.1	14.6	13.5	12.6	11.0	9.0	7.7	8.0	10.1	11.5	14.3	15.0
	5% WB	MCDBR	16.5	17.9	17.4	19.2	18.6	16.0	14.9	13.5	13.2	13.6	14.7	14.8
		MCWBR	16.1	15.6	14.0	12.6	11.0	8.6	7.5	7.2	8.9	10.9	14.1	15.5
(40) Clear-Sky Solar Irradiance	taub		0.309	0.318	0.341	0.377	0.427	0.471	0.489	0.462	0.390	0.349	0.319	0.307
	taud		2.474	2.454	2.440	2.371	2.262	2.176	2.138	2.204	2.378	2.508	2.544	2.539
	Ebn at Noon		269	284	288	284	271	258	252	255	268	269	264	261
	Edh at Noon		24	29	32	37	43	46	48	43	34	27	23	21
(44) All-Sky Solar Radiation	RadAvg		587	863	1181	1525	1703	1857	1894	1685	1384	935	652	490
	RadStd		47	79	94	161	168	180	103	101	113	89	53	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+104

Nomenclature: See separate page